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Fabricating Organic Nanotubes through Selective Disassembly of Two-Dimensional Covalent Organic Frameworks

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Supporting Information

ABSTRACT: Covalent organic frameworks (COFs) are an emerging class of crystalline porous organic polymers with potential for innovative applications. Here we report the use of COFs as precursors for the fabrication of well-defined tubular nanomaterials. A proof-of-concept study is presented for the controllable fabrication of organic nanotubes through selective disassembly of two-dimensional heteropore COFs. Two dual-pore COFs are constructed based on orthogonal reactions. Each COF possesses two different kinds of pores, which are formed by linking all-hydrzaone-bonded nanopores with boroxines. Selectively hydrolyzing boroxine rings in the COFs while keeping hydrazone linkages untouched gives rise to organic nanotubes with diameters and shapes corresponding to the nanochannels of the COFs.

Covalent organic frameworks (COFs) are a class of two-dimensional (2D) or three-dimensional (3D) crystalline porous organic polymers constructed by linking building blocks with covalent bonds (linkages).[1] A distinguishing feature of COFs that makes them differ from amorphous porous organic polymers is the highly organized channel structures in their frameworks. Most importantly, apertures of channels in COFs, usually ranging from several angstroms to several nanometers, can be precisely adjusted at the atomic level by varying the sizes of the building blocks, whereas their shapes can be controlled through the choice of the symmetry of the building blocks.[2] Benefiting from the well-ordered channel structures, COFs offer versatile applications such as materials adsorption and storage,[3] separation,[4] sensing,[5] catalysis,[6] and drug delivery.[7] For these applications, the channels serve their functions as a whole, and thus COFs are generally regarded as porous materials containing well-ordered nanochannels.

Despite the significant achievements made over the past decade, exploring new applications of COFs is still a key aspect in this field. However, so far most of the attention has been focused on exploiting the porosity and conjugated backbones of COFs for applications in adsorption, storage, separation, photo/electronics, and so on.[8] We herein report the development of an innovative application of 2D COFs based on an understanding of 2D COFs from a different perspective. Unlike the channels in 3D COFs, which are interpenetrated each other, the channels in 2D COFs are generated from stacking of porous 2D layers and thus exist as independent one-dimensional (1D) conduits. In this context, 2D COFs can be regarded as orderly bundled nanotube arrays. Therefore, detaching the bundled nanotubes via controllable disassembly of 2D COFs may be a new approach to produce organic nanotubes, a large class of tubular nanomaterials that have drawn considerable attention since the discovery of carbon nanotubes in 1991,[9] but their preparation with precisely controlled diameter, shape, and functionality is still challenging and mainly relies on a bottom-up strategy.[10] Thanks to the crystalline structures of COFs, one advantage of this proposed strategy is that both the diameter and the shape of nanotubes can be precisely designed, predicted, and controlled, since they are inherited from their COF precursors. Different from the bottom-up approach, as represented by stacked assembly of macrocycles,[11] the present strategy is a top-down approach which nicely complements the previous one.

The implementation of this approach relies on selectively breaking target covalent bonds in COF precursors while maintaining the 1D channel structures so as to isolate individual nanotubes. Apparently, COFs with homogeneous porosity do not work for this idea because they are composed of all the same kind of pores, and thus bond-breaking will lead to deconstruction of the channels in the COFs. By contrast, 2D heteropore COFs, an emerging type of COFs in which different kinds of pores are integrated into one framework, may meet this requirement due to their heterogeneous pore structures.[12] This proof-of-concept study is illustrated in Scheme 1. A dual-pore COF is designed to be constructed through orthogonal reactions,[13] here hydrazone formation and trimerization of boronic acids. Hydrazone linkage has excellent stability against hydrolysis, while boroxine is apt to be hydrolyzed. Their dramatic difference in hydrolytic stability is favored to accomplish selective disassembly of the heteropore COF by completely hydrolyzing the boroxine rings while keeping the hydrazone units unaffected. In this context, the nanochannels, which are formed by the stacked hydrazone-linked macrocycles in the COF, can be maintained and detached from each other to produce individual organic nanotubes with the diameter and shape corresponding to the nanochannels in the precursor COF.

Directed by the above design, a dual-pore COF carrying ethyoxyl groups (**COF-OEt**) was synthesized by solvothermal

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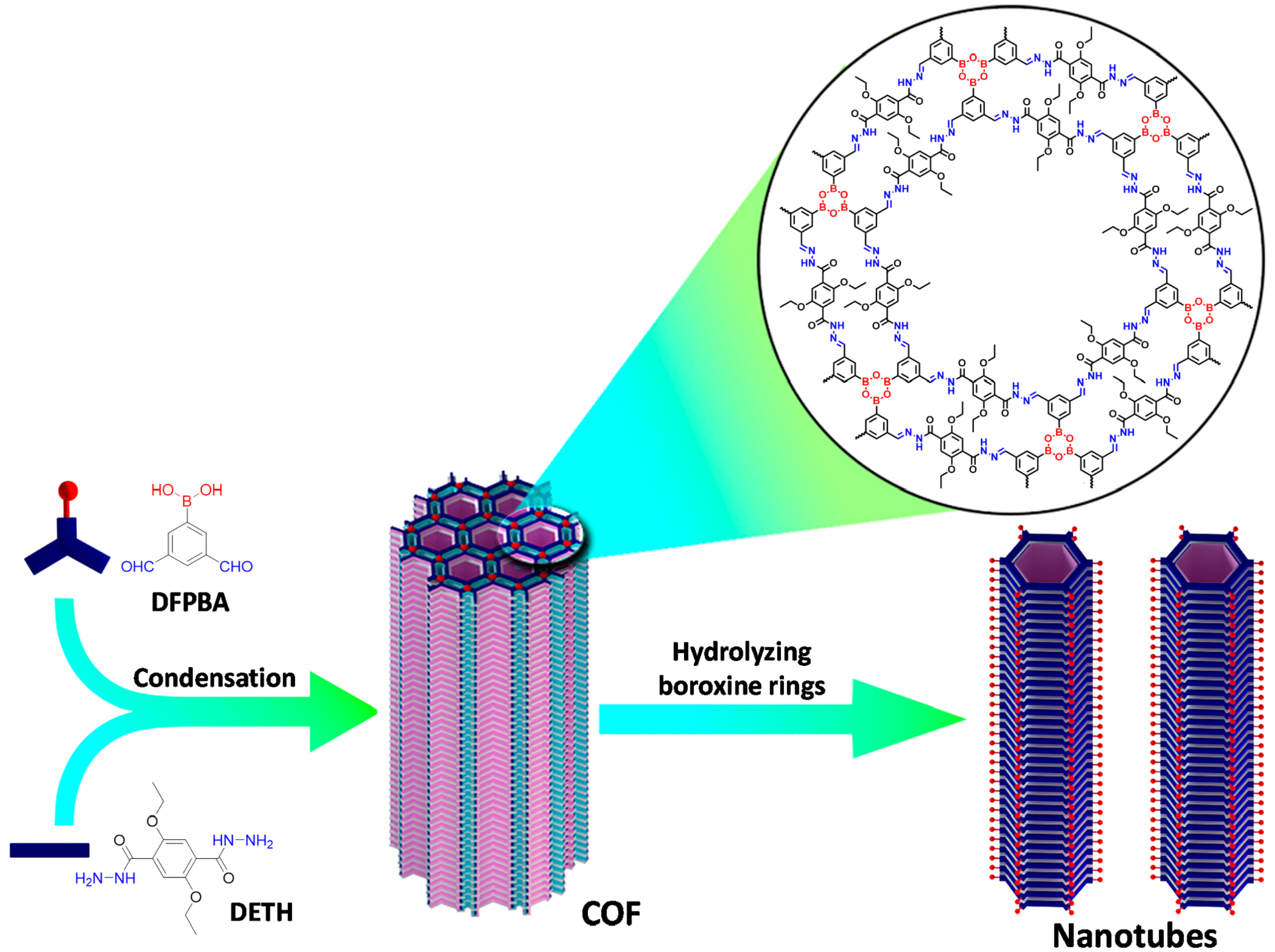
Scheme 1. Schematic Illustration for Fabrication of Nanotubes through Selectively Hydrolyzing a 2D Heteropore COF



reaction of 3,5-diformylphenylboronic acid (DFPBA) and 2,5-diethoxyterephthalohydrazide (DETH) (Scheme 1). Interestingly, different from most COFs which are produced as powders under solvothermal conditions, **COF-OEt** was initially obtained as a colloid after the condensation (Figure S1a), which is a rare phenomenon for COFs and is beneficial for processing of COFs.[14] The Fourier transform infrared (FT-IR) spectrum of the as-obtained yellow crystallites shows the characteristic bands of C═N moieties and B_3O_3, accompanied by significant attenuations in the intensities of −CHO, −OH, and $-NHNH_2$ bands in comparison with those of the monomers (Figure S2), indicating formation of hydrazone bonds and boroxines. Solid-state ^{13}C cross-polarization with magic angle spinning (CP-MAS) NMR spectrum of **COF-OEt** further supports the formation of hydrazone bonds, as evidenced by the signal at 148.9 ppm which is corresponding to the carbon atom of hydrazone units (Figure S3). As revealed by thermogravimetric analysis (TGA), **COF-OEt** shows good thermal stability (Figure S4). Scanning electron microscopy (SEM) reveals that the crystallites aggregate into irregular particles (Figure S5).

Powder X-ray diffraction (PXRD) measurement confirmed the formation of the predicted crystalline framework structure of **COF-OEt**. Its experimental PXRD profile shows an intense diffraction peak at 2.42°, as well as a series of weaker peaks at 4.05°, 6.59°, 8.85°, 11.50°, and 22.27°, respectively (Figure 1a). To elucidate its crystal structure, simulated PXRD patterns were generated based on the predicted structural models with eclipsed (AA) stacking (Figure 1c) and staggered (AB) stacking (Figure 1d) using Materials Studio software.

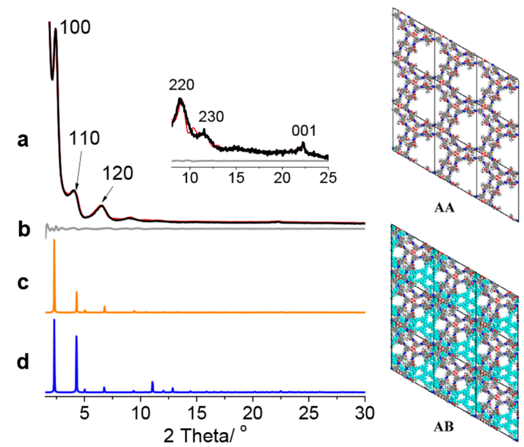


Figure 1. Left: (a) Experimental (black) and refined (red) PXRD patterns of **COF-OEt**. (b) Difference plot between the experimental and refined PXRD patterns. Simulated PXRD patterns of (c) AA and (d) AB stacking models. Right: Illustrations of AA and AB stacking models.

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The comparison between the experimental PXRD data and the simulated PXRD patterns indicates that the experimental PXRD profile matches well with the stimulated one with AA stacking. The lower total energy of the model with AA stacking (313.5 kcal/mol) than that of AB stacking (606.8 kcal/mol) further supports that **COF-OEt** adopts AA stacking. Pawley refinement was performed to give the unit cell parameters of $a = b = 38.55$ Å, $c = 4.07$ Å, $\alpha = \beta = 90°$, and $\gamma = 120°$, with $R_p = 2.52\%$, and $R_{wp} = 3.54\%$. The difference plot suggests that the refined PXRD pattern matches well with the experimental result (Figure 1b).

Nitrogen adsorption−desorption measurement was carried out to assess the porosity of **COF-OEt**. Its Brunauer−Emmett−Teller (BET) surface area was calculated from its nitrogen adsorption isotherm at 77 K (Figure 2a) to be 465 m^2/g (Figure S6a) and a total pore volume (at $P/P_o = 0.99$) of 0.85 cm^3/g was obtained. The relatively low surface area might be attributed to the gelation after the condensation reaction. Due to the strong interactions between frameworks and solvents, solvent molecules are extremely hard to be removed from the COF. Pore size distribution (PSD) analysis reveals a main distribution around 23.5 Å (Figure 2b), which is in good agreement with the theoretical pore aperture of the hexagonal pores (21.1 Å). However, the rectangular pores could not be identified in the PSD profile as their width (2.67 Å) is too narrow to accommodate N_2 molecules which have a kinetic diameter of 3.64 Å.[15] This phenomenon was also previously observed in other COFs.[13d,16]

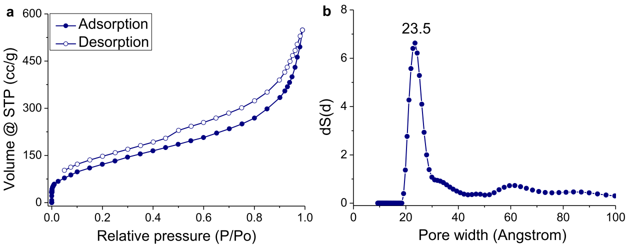

Figure 2. (a) N_2 adsorption−desorption isotherm at 77 K. (b) Pore size distribution profile of **COF-OEt**.

With the target dual-pore COF in hand, the design of fabricating organic nanotubes from controllable disassembly of the COFs was implemented in the next step. The experiment was carried out by suspending **COF-OEt** in an aqueous hydrochloric acid solution (6 M) for several days at room temperature. IR spectrum of the hydrolyzate reveals that boroxine rings in the COF are completely hydrolyzed while the hydrazone units are still presented (Figure S7). The hydrolysis product was further visualized with transmission electron microscopy (TEM). As shown in Figure 3, the pristine COF displays a lamellar morphology. However, after the hydrolysis, hollow tubular morphology is observed, indicating the COF is disassembled to afford organic nanotubes (named **NT-OEt**). Most of the nanotubes intertwine with each other (Figure 3b and Figure S8), which should be driven by the hydrogen bonding between boronic acid groups located at the peripheral of the nanotubes. Fortunately, isolated nanotubes could be identified, with their mean diameter to be 5.5 (±0.6) nm, which is in agreement with the theoretical outer diameter of the nanotube estimated from the precursor COF (4.3 nm, Figure S9a). To evaluate the stability of **NT-OEt**, the sample was sonicated for 10 min, after which the nanotubes

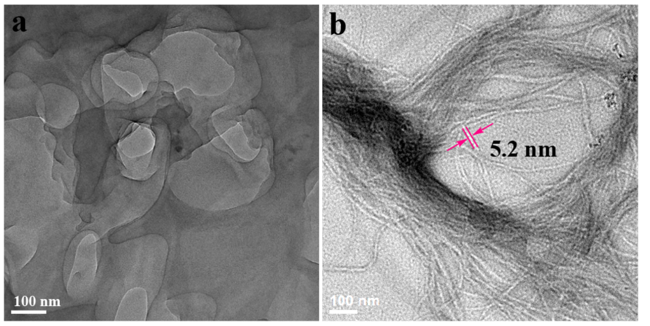

Figure 3. TEM images of **COF-OEt** before (a) and after (b) hydrolysis.

disappeared (Figure S10). This result suggests that the as-formed organic nanotubes are labile upon ultrasonic treatment due to the weak stacking interactions between macrocycles. The resulting suspension after the ultrasonic treatment was stood for 12 h at room temperature. No regeneration of nanotubes was observed by TEM (Figure S11), indicating that the nanotubes originated directly from the channels of the COF precursor.

To improve stability of the nanotubes, a practical method is to covalently link the stacked macrocycles.[11a] To this end, a COF similar to **COF-OEt** but bearing allyloxy groups (named **COF-OAl**) was designed and synthesized from the condensation of DFPBA and 2,5-bis(allyloxy)terephthalohydrazide (BATH) (Scheme S1). Its chemical composition was characterized with FT-IR and solid state CP/MAS ^{13}C NMR spectroscopies and the experimental data confirmed the presence of hydrazone bonds and boroxines in the COF (Figures S12 and S13). Its dual-pore structure was corroborated to be quite similar to that of **COF-OEt**, with lattice parameters of $a = b = 39.00$ Å, $c = 4.07$ Å, $\alpha = \beta = 90°$, and $\gamma = 120°$, with $R_{wp} = 2.59\%$, $R_p = 1.80\%$ (Figure S14). Nitrogen adsorption−desorption measurement (Figure S15a) confirmed the porosity of **COF-OAl** with a low BET surface area of 202 m^2/g due to the gelation (Figures S1b and S6b). A main distribution around 20.0 Å appears in its PSD profile (Figure S15b), matching well with the theoretical diameter of the hexagonal pores (19.5 Å). All the results provide compelling evidence for the formation of the designed dual-pore COF.

Interlayer cross-linking in **COF-OAl** was performed through alkene metathesis reaction of allyloxy groups (the product after cross-linking was termed **COF-OAl-CR**).[17] Its experimental PXRD profile reveals retention of the crystalline structure of **COF-OAl** (Figure S16). Its CP/MAS ^{13}C NMR and FT-IR spectra are similar to that of **COF-OAl**, except for significant decreases in the intensity of $=CH_2$ bands around 3077 and 2960 cm^{-1} in the IR spectrum of **COF-OAl-CR** (Figures S17 and S18), which could be attributed to the consumption of $=CH_2$ by the metathesis reaction. After the cross-linking reaction, the resulting **COF-OAl-CR** was further hydrolyzed. The significant attenuation of B_3O_3 in IR spectrum of the hydrolyzate indicates hydrolysis of boroxine rings in the COF, while the hydrazone units are still reserved as the C=N band remains unchanged (Figure S19). The product after the hydrolysis was observed with TEM, which indicated the existence of nanotubes (named **NT-OAl-CR**). By contrast, the precursor COFs exhibit lamellar morphology (Figure S20). Similar to **NT-OEt**, the as-prepared nanotubes aggregate

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together (Figure S21). The mean diameter of the nanotubes was obtained from the isolated nanotubes to be 5.6 (±0.5) nm (Figure 4a), matching with the predicted theoretical size (4.2 nm, Figure S9b). To verify its stability, the as-prepared **NT-OAl-CR** was treated under ultrasound for 10 min and then submitted to TEM observation again. It is found that nanotubes still exist (Figure 4b), indicating that the cross-linking dramatically improves stability of the organic nanotubes. Moreover, intertwining of two or three nanotubes could be clearly observed.

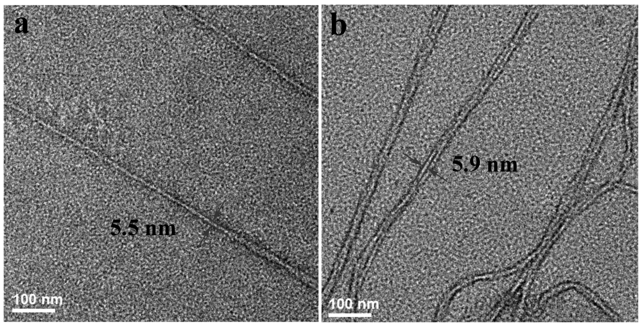

Figure 4. TEM images of **NT-OAl-CR** before (a) and after (b) ultrasonic treatment.

In conclusion, taking advantage of the differences in pore structures and chemical stability of linkages in heteropore COFs, organic nanotubes were fabricated through selectively hydrolyzing one type of linkages while keeping the other linkages untouched. In this way, an innovative application has been developed for COFs. Thanks to the pre-designable and highly ordered crystalline structures of COFs, this approach offers designable and precise fabrication of organic nanotubes. It may be a general top-down approach because use of two different kinds of linkages is not a must, provided linkages exhibiting different stability in different kinds of pores.[18] Furthermore, functionalization of nanotubes can be realized through modification on channel surfaces of COFs. This work well demonstrates an exclusive application of heteropore COFs originating from their unique structure features, which differ from the COFs with homogeneous pore structures. On the other hand, this strategy should also be able to be used to synthesize macrocycles which are difficult to prepare or even inaccessible under de novo synthetic conditions. Such potential is currently under investigation in our laboratory.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.9b11401.

Procedures for the preparation of the COFs, IR spectra, solid-state ^{13}C CP-MAS NMR spectra, SEM and TEM images, BET plots, TGA traces, additional PXRD profiles, and fractional atomic coordinates for the unit cells of the COFs, including Figures S1−S24, Tables S1 and S2, and Scheme S1 (PDF)

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Notes

The authors declare no competing financial interest.

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(17) According to the mechanism of alkene metathesis reaction and the crystal structure of **COF-OAl**, interchannel cross-linking is unfavorable (Figure S22).

(18) For example, an imine bond in different environments exhibits different chemical stability. For references, see: (a) Xu, H.; Gao, J.; Jiang, D. Stable, crystalline, porous, covalent organic frameworks as a platform for chiral organocatalysts. *Nat. Chem.* **2015**, *7*, 905−912. (b) Kandambeth, S.; Shinde, D. B.; Panda, M. K.; Lukose, B.; Heine, T.; Banerjee, R. Enhancement of chemical stability and crystallinity in porphyrin-containing covalent organic frameworks by intramolecular hydrogen bonds. *Angew. Chem., Int. Ed.* **2013**, *52*, 13052−13056.

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